

Research Proposal

Uncertainty-Guided Graph-Aware Mixture-of-Experts for Multimodal Microservice Anomaly Detection

1 1. Research Objective

The objective of this research is to develop a robust multimodal anomaly detection framework for microservice systems by jointly exploiting logs, traces, and metrics.

The proposed framework is designed to classify each observation as:

Normal or Anomalous

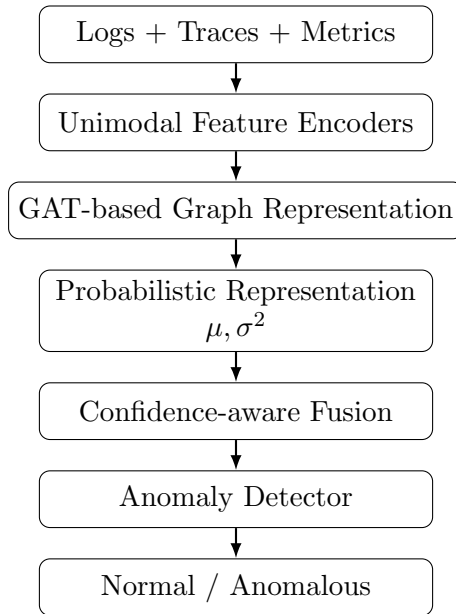
The research focuses exclusively on multimodal anomaly detection. Root-cause analysis, root-cause ranking, failure attribution, and counterfactual reasoning are outside the scope of the proposed work.

2 2. Base Research Framework

The base framework considered in this research is MUAD, an uncertainty-aware multimodal anomaly detection framework for microservice systems.

MUAD jointly exploits logs, traces, and metrics and addresses both intra-modal and inter-modal uncertainty. Its architecture consists of unimodal feature representation, a Graph-based Probabilistic Encoder (GPE), a Confidence-aware Fusion Mechanism (CFM), and an anomaly detector.

The base workflow can be represented as:



3 3. Research Gap and Proposed Modification

The base framework employs a GAT-based graph representation within the Graph-based Probabilistic Encoder. While this provides dependency-aware representation learning, the architecture uses a single graph-attention mechanism for representation extraction.

The proposed research investigates whether a dynamically routed Mixture-of-Experts architecture can provide more specialized representations for heterogeneous multimodal observations.

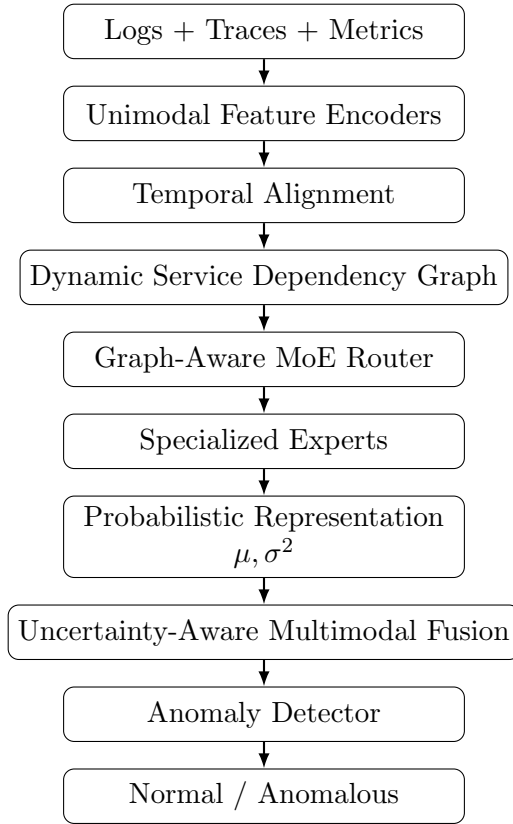
The principal architectural modification is therefore:

$$\boxed{\text{GAT} \longrightarrow \text{Graph-Aware Mixture-of-Experts}}$$

The proposed framework retains the multimodal setting and probabilistic uncertainty representation of the base framework while introducing dynamic expert specialization and uncertainty-guided routing.

4 4. Proposed Framework

The proposed workflow is:



5 5. Specialized Expert Architecture

The proposed Mixture-of-Experts layer consists of four specialized experts:

Table 1: Specialized experts in the proposed framework

Expert	Functional role
Temporal Expert	Models temporal evolution, trends, abrupt changes, and temporal dependencies.
Semantic Expert	Captures semantic and event-level information, particularly from log observations.
Dependency Expert	Models relationships among microservices using the service dependency graph.
Cross-Modal Expert	Learns complementary interactions among logs, traces, and metrics.

The expert representation is defined as:

$$H_v = \sum_{k=1}^K g_{v,k} E_{v,k}(X_v, A), \quad (1)$$

where $E_{v,k}$ denotes expert k , $g_{v,k}$ denotes the learned routing weight, X_v denotes the modality representation, and A represents the service dependency graph.

6. Graph-Aware Expert Routing

The expert router receives both multimodal representations and graph information:

$$g_v = Router(X_v, A, U_v), \quad (2)$$

where U_v represents uncertainty associated with modality v .

The normalized routing weights are:

$$g_{v,k} = \frac{\exp(q_{v,k})}{\sum_{j=1}^K \exp(q_{v,j})}. \quad (3)$$

Consequently, expert selection is data-dependent rather than fixed.

7. Probabilistic Representation

The proposed framework retains probabilistic representation learning from the base framework.

For each modality:

$$\mu_v, \sigma_v^2 = MLP(H_v). \quad (4)$$

The latent representation is sampled as:

$$z_v = \mu_v + \sigma_v \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (5)$$

A KL-divergence regularization term can be used:

$$\mathcal{L}_{KL} = \frac{1}{3} \sum_v KL [\mathcal{N}(\mu_v, \sigma_v^2) \| \mathcal{N}(0, I)]. \quad (6)$$

8 8. Uncertainty-Aware Multimodal Fusion

Let:

$$z_L, \quad z_T, \quad z_M$$

represent the log, trace, and metric latent representations.

A modality-specific reliability score is estimated by:

$$c_v = MLP(z_v, U_v). \quad (7)$$

The normalized modality weights are:

$$\omega_v = \frac{\exp(c_v)}{\sum_j \exp(c_j)}. \quad (8)$$

The fused representation is:

$$z_f = \omega_L z_L + \omega_T z_T + \omega_M z_M. \quad (9)$$

The final anomaly prediction is:

$$\hat{y} = Softmax(W z_f + b). \quad (10)$$

9 9. Research Contributions

The proposed research will investigate the following contributions:

1. **Graph-Aware Mixture-of-Experts:** A dynamic expert representation mechanism that replaces the single GAT-based representation with specialized temporal, semantic, dependency, and cross-modal experts.
2. **Uncertainty-Guided Expert Routing:** An expert-routing mechanism that incorporates uncertainty into the selection of specialized experts.
3. **Probabilistic Multimodal Representation:** A unified probabilistic representation of heterogeneous log, trace, and metric features.
4. **Uncertainty-Aware Multimodal Fusion:** An adaptive fusion mechanism that dynamically adjusts the contribution of each modality according to its estimated reliability.

10 10. Experimental Research Questions

- RQ1:** Does the proposed Graph-Aware MoE improve multimodal anomaly detection compared with the GAT-based representation?
- RQ2:** Does dynamic expert routing provide additional benefit over a single shared representation?
- RQ3:** Does uncertainty-guided routing improve robustness under noisy or ambiguous observations?
- RQ4:** Does uncertainty-aware multimodal fusion improve anomaly detection over direct feature concatenation or fixed-weight fusion?

RQ5: How robust is the proposed framework when one or more modalities are partially unavailable?

RQ6: How sensitive is the proposed framework to its principal hyperparameters?

11. Baseline and Comparative Evaluation

The following models should be evaluated:

Table 2: Comparative experimental configurations

Configuration	Purpose
Base GAT Model	Evaluate the original graph-attention representation.
Plain MoE	Evaluate expert specialization without graph information.
Graph-MoE	Evaluate graph-aware expert routing without uncertainty guidance.
Graph-MoE + Probabilistic Representation	Evaluate the contribution of uncertainty modeling.
Full Proposed Model	Evaluate graph-aware MoE, uncertainty-guided routing, probabilistic representation, and adaptive fusion jointly.

12. Ablation Study

The following ablations will be performed:

Table 3: Ablation configurations

Variant	Modification
w/o Graph	Remove service dependency information.
w/o MoE	Replace the expert mixture with a single shared representation.
w/o Dynamic Routing	Use fixed or equal expert weights.
w/o Uncertainty Routing	Remove uncertainty from expert routing.
w/o Probabilistic Representation	Use deterministic embeddings instead of (μ, σ^2) .
w/o Adaptive Fusion	Concatenate modality representations directly.

These experiments will isolate the contribution of each proposed component.

13. Modality-Level Evaluation

The contribution of multimodal information will be examined using:

Table 4: Modality combinations

Configuration	Modalities
Metric only	M
Log only	L
Trace only	T
Log + Metric	L+M
Log + Trace	L+T
Trace + Metric	T+M
Log + Trace + Metric	L+T+M

For each configuration, Precision, Recall, F1-score, and PR-AUC will be reported.

14. Robustness Evaluation

Two classes of robustness experiments will be conducted.

14.1 Missing Modality

The model will be evaluated with:

0%, 10%, 20%, 40%, 60%

missing observations.

The missing information will be introduced independently for logs, traces, and metrics as well as through random multimodal masking.

14.2 Modality Noise

Controlled noise will be introduced into one or more modalities.

Examples include:

Clean Log + Noisy Metric + Clean Trace

Noisy Log + Clean Metric + Clean Trace

Noisy Log + Noisy Metric + Clean Trace.

Performance degradation will be measured as:

$$Degradation(\%) = \frac{F1_{\text{clean}} - F1_{\text{perturbed}}}{F1_{\text{clean}}} \times 100. \quad (11)$$

15. Expert Routing Analysis

The learned routing weights will be analyzed for different anomaly conditions.

For each sample:

$$[g_1, g_2, \dots, g_K]$$

will be recorded.

The average expert contribution is:

$$\bar{g}_k = \frac{1}{N} \sum_{i=1}^N g_{i,k}. \quad (12)$$

A heatmap of:

Anomaly Type $\times$ Expert Weight

will be generated to determine whether the experts develop differentiated specializations.

16 16. Uncertainty Analysis

For each modality, the average uncertainty will be estimated by:

$$U_v = \frac{1}{D} \sum_{d=1}^D \sigma_{v,d}^2. \quad (13)$$

The analysis will include:

- modality uncertainty,
- expert-routing entropy,
- modality fusion weights,
- prediction confidence.

The routing entropy is defined as:

$$H_g = - \sum_{k=1}^K g_k \log g_k. \quad (14)$$

17 17. Hyperparameter Sensitivity

The following parameters will be systematically varied.

17.1 Number of Experts

$$K \in \{2, 3, 4, 5, 6\}.$$

17.2 Embedding Dimension

$$D \in \{32, 64, 128, 256\}.$$

17.3 Temporal Window Size

$$T \in \{5, 10, 20, 30\}.$$

17.4 Uncertainty Loss Weight

$$\lambda_{KL} \in \{0.001, 0.01, 0.1, 0.5, 1.0\}.$$

17.5 Learning Rate

$$lr \in \{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}.$$

For each configuration, Precision, Recall, F1-score, PR-AUC, and computational cost will be recorded where appropriate.

18 18. Results to be Reported

The final manuscript will include the following result categories.

18.1 Overall Performance

Table 5: Overall anomaly detection performance				
Model	Precision	Recall	F1	PR-AUC
Base GAT	—	—	—	—
Plain MoE	—	—	—	—
Graph-MoE	—	—	—	—
Proposed Model	—	—	—	—

18.2 Ablation Results

Table 6: Component-level ablation results				
Variant	Precision	Recall	F1	PR-AUC
Full Model	—	—	—	—
w/o Graph	—	—	—	—
w/o MoE	—	—	—	—
w/o Dynamic Routing	—	—	—	—
w/o Uncertainty Routing	—	—	—	—
w/o Probabilistic Representation	—	—	—	—
w/o Adaptive Fusion	—	—	—	—

No numerical results should be inserted until the corresponding experiments have been completed.

19 19. Experimental Evidence for the Contributions

Table 7: Mapping between contributions and required evidence

Contribution	Required evidence
Graph-Aware MoE	Comparison of GAT, plain MoE, and Graph-MoE.
Dynamic Expert Routing	Comparison of learned routing and fixed/equal routing, together with expert-weight analysis.
Uncertainty-Guided Routing	Comparison of deterministic routing and uncertainty-guided routing under clean and degraded observations.
Probabilistic Representation	Comparison between probabilistic and deterministic representations.
Adaptive Multimodal Fusion	Comparison of concatenation, fixed weighting, and uncertainty-aware fusion.
Robustness	Controlled missing-modality and noise experiments.
Hyperparameter Stability	Sensitivity analysis across the major architectural and training parameters.

20 20. Statistical Evaluation

The major experiments should be repeated using multiple random seeds.

The reported results should preferably be presented as:

$$\text{mean} \pm \text{standard deviation.}$$

The final comparison should consider both absolute performance and performance degradation under controlled perturbations.

21 21. Implementation Plan

The implementation will proceed sequentially:

1. Prepare and align logs, traces, and metrics.
2. Construct the service dependency graph.
3. Implement the unimodal feature encoders.
4. Reproduce the GAT-based baseline.
5. Implement the four specialized experts.
6. Implement graph-aware MoE routing.
7. Add probabilistic representation learning.
8. Add uncertainty-guided routing.
9. Implement uncertainty-aware multimodal fusion.
10. Train and evaluate the complete model.
11. Conduct ablation experiments.

12. Conduct robustness experiments.
13. Perform expert-routing and uncertainty analysis.
14. Perform hyperparameter sensitivity experiments.

22 22. Final Research Scope

The proposed research is limited to:

Multimodal Anomaly Detection

using:

Logs + Traces + Metrics

with the principal methodological contribution:

GAT $\rightarrow$ Graph-Aware Uncertainty-Guided MoE

The final model produces:

Normal / Anomalous

and evaluates detection performance, robustness, expert behavior, uncertainty modeling, and hyperparameter sensitivity.

Reference

H. Dou, S. Guo, L. Zou, Y. Zhang, P. Chen, and Z. Zheng, “Uncertainty-Aware Multimodal Anomaly Detection for Microservice Systems With Active Learning,” *IEEE Transactions on Services Computing*, vol. 19, no. 3, 2026, doi: 10.1109/TSC.2026.3672587.